Consider Gridworld, which has four primitive actions of up, right, left, and down, and each action lasts one time step. In the options framework, there are options rather than just primitive actions. An option is the combination of an *option policy* (which like a regular policy takes a state and returns a probability distribution over actions), a *termination condition*, and an *input set* (which is a subset of states). The idea is that a particular option gets triggered when the agent encounters a state in the option's input set, and that particular option's policy is run until the termination condition is met, at which point a different option may be selected. These option policies might be simpler policies than a single, big, deep neural network policy that we have implemented in this book. But by intelligently selecting these higher-level options, efficiencies can be gained by not having to use a more computationally intensive policy for taking each primitive step.

11.2.4. Model-based planning

We already discussed the idea of *models* in reinforcement learning in two contexts. In the first, a model is simply another term for an approximating function like a neural network. We sometimes just refer to our neural network as a model, since it approximates or models the value function or the policy function.

The other context is when we refer to *model-based* versus *model-free learning*. In both cases we are using a neural network as a model of the value function or a policy, but in this case model-based means the agent is making decisions based on an explicitly constructed model of the dynamics of the environment itself rather than just its value function. In model-free learning, all we care about is learning to accurately predict rewards, which may or may not require a deep understanding of how the environment actually works. In model-based learning, we actually want to learn how the environment works. Metaphorically, in model-free learning we are satisfied knowing that there is something called gravity that makes things fall, and we make use of this phenomenon, but in model-based learning we want to actually approximate the laws of gravity.

Our model-free DQN worked surprisingly well, especially when combined with other advances like curiosity, so what is the advantage of explicitly